

A Study of Light Intensity of Stars for Exoplanet Detection using Machine Learning

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Abstract—Exoplanets is defined as the planets outside the solar system. With a hope to find a habitable, earth-like planet, the detection of exoplanet is very crucial domain of research. Many researchers and space agencies are working on manifold variation of strategies to find novel exoplanets. In this paper, we explore the detection of exoplanets using machine learning classification. We explore time-series based light intensity data of different stars to understand a pattern that correlates with presence of exoplanets in that star's orbit. The data used in this study was collected under the NASA Kepler mission - Campaign 3. After initial analysis of the said data, we further implemented various classification algorithm to detect exoplanets. The K-nearest neighbor model trained on SMOTE re-balanced data gave an accuracy of 98.20% with a F1-score of 98%. Further, understanding the difficulty of collecting data in real time, we considered implementing the classification models on limited data by limiting the observations of light intensity for each star to multiple thresholds between 75% to 1%. The accuracy and F1-score for even the model with only 1% data for KNN model was found to be 90.1% and 90% respectively.

Keywords — Exoplanets, Classification, KNN, SMOTE and Kepler mission.

I. INTRODUCTION

The research in the domain of astronomy and space science has been growing exponentially, especially in regard to the generation of relevant data [1] [2]. A crucial reason for such advancements is due to research in machine learning which like healthcare [21] [23], logistics [22], education [20] and business [17] is now also being used for astronomy. With an easy infrastructure and policies to collect, maintain and distribute data, there is tremendous potential for machine learning-based systems to be implemented for the analysis and development of frameworks for space science research [1] [3] [4] [5] [14] [15]. In this paper, we particularly focus on exoplanets detection. Exoplanets are often defined as the undiscovered planets outside our solar system. Scientist believe that there might be certain planets orbiting around star other than sun, that might have natural condition to be habitable for life. But in order to find such habitable planets, the primary objective is to find new exoplanets in the universe. Researchers have come up with multiple methods to do the same, which has

been discussed in details in the next section. However, with growing availability of data, we considered using machine learning with an aim to create robust method than the methods in the literature. Particularly, we explore a time-series data that observes the light intensity of various stars over a period of time. Planets do not emit light of their own, but the stars that they orbit around to do. If a star has fluctuating light intensity over a period of time, then the plausible reason is that there is a planet orbiting that star. Thus, if an observation of light intensity in a fluctuating pattern is found over a period of time for a particular star, then the star is a potential one with the presence of an exoplanets. This paper aims to answer two main questions:

- Can machine learning be used to detect exoplanets by studying the light intensity of stars?
- If so, can the same be optimally implemented even if a limited observation of light intensity is available?

II. BACKGROUND AND DATA

A. Literature study

Baron in [6] highlights certain key developments in research that uses machine learning for astronomy. Baron mentions that the most widely used ML algorithms in astronomy are in a supervised setting and also highlights the potential of the unsupervised algorithms in specifically extracting hidden knowledge from the massive astronomical data. In [16], author analyses the orbital parameter using Machine Learning for live tracking of a satellite. Similarly in [7], Fluke et. al. mentioned seven main categories of implementation of ML on astronomical datasets: classification, regression, clustering, forecasting, generation, discovery, and the development of new scientific insight. In particular, the research in exoplanet detection is not new. There has been a manifold variation of research that has been done around the same domain based on various scientific methods. Wright [8] used data features like radial velocities, astrometry, direct imaging, transits, and gravitational microlensing to detect planetary companions to stars from an observational perspective. In [12], Akenson et. al. highlights the tools and content for exoplanet research

at NASA. The current literature consists of interactive tables containing properties of all published exoplanets, Kepler planet candidates, threshold-crossing events, data validation reports and target stellar parameters, light curves from the Kepler and CoRoT missions and from several ground-based surveys, and spectra and radial velocity measurements from the literature. The research on exoplanet is not just limited to detection but far beyond that. In [13], Zhao et. al. characterizes the interiors of a gaseous giant exoplanet to help understand planet formation in diverse protoplanetary environments. However, there is always a scope for advancements. The detection of exoplanets outside the solar system has been of great scope in the last decade, with successfully proven discoveries using multiple methods. Specifically, with more accessibility to data and ease of ML implementation, in this paper, we try to use ML algorithms in detecting exoplanets in a novel way. Perryman [19] has provided a detailed explanation of exoplanet, their characteristics with broad-range of parameter based on evidence from last 25 years.

B. Data

The dataset used in this research is openly available and has been derived from observations made by the NASA Kepler space telescope. Over 99% of this dataset originates from Campaign 3 of the Kepler mission. We title this dataset “Kepler Exoplanet Detection Campaign 3” (KEDC3) data [9]. The dataset is divided into two parts namely train and test. The train set consists of 5087 records of stars. Out of these 37 are confirmed exoplanet stars (1) and 5050 are labeled as non-exoplanet stars (0). For each star, there are 3198 observations of light intensity (flux) over a time period. While the test set consists of 570 rows with 5 confirmed exoplanet stars.

III. PRIOR ANALYSIS - PATTERN OF DATA

Before implementing machine learning algorithms over the data to make predictions, a prior analysis of the data was done. In figure 1, we considered 5 randomly sampled stars of both categories; without exoplanets (a) and with exoplanets (b). Further, we plotted the observation of light intensities over time. As it can be seen from the figure, the stars with exoplanets tend to have a cyclic structure with a repeating pattern of flux after a frequency. Whereas, stars with no exoplanets don’t have a longitudinal repeating pattern though has sudden spikes in light intensity. This preliminarily confirms that there is unique behavior that stars with exoplanets show.

Furthermore, we considered the Gaussian histogram of flux values for 3 different stars with exoplanets and without exoplanets (Fig 2). The interpretation of this analysis adds to the previously confirmed preliminary results. Even here, the difference between patterns of both classes can be clearly observed. The flux values for stars without exoplanets have a limited range of values with greater frequency of occurrence whereas for flux values for stars with exoplanets there is a greater range of values with lesser frequency each. This

confirms the fluctuating nature of stars that has a planet in their orbit. Although, this analysis could establish a ground to differentiate patterns of both the classes but the end goal of this experiment is to develop a machine learning model that replicates this trend.

IV. PREDICTION OF EXOPLANETS

Further, we implemented various machine learning algorithms on the training data and tested the model on the test data to evaluate its performance. The machine learning algorithms that we considered are logistics regression, K-nearest neighbor (KNN), and Decision tree. Logistic regression is based on sigmoid function applied over a linear predictor. It is one of the popularly used classifier algorithm. KNN is a probability distribution method that agglomerates ‘k’ closest training examples in a featured space, identifying as single class. Decision tree is tree data-structure based classifier that derives rules for classification [10] [24]. Further, since the dataset was highly imbalanced, we considered using Synthetic Minority Oversampling Technique (SMOTE) [11] to re-balance the classes in the KEDC3 data. It simply over samples the observations in the minority class. This method is optimal since the new generated observations are relatively close in feature space to existing examples from the minority class.

TABLE I
PERFORMANCE OF THE IMPLEMENTED MODELS FOR ORIGINAL AND
BALANCED DATA

	Accuracy	Precision	Recall	F1-Score
Original Data				
LR	0.991	[0.99, 0]	[1, 0]	[1, 0]
KNN	0.991	[0.99, 0]	[1, 0]	[1, 0]
Decision tree	0.989	[0.99, 0.33]	[1, 0.2]	[0.99, 0.25]
Balanced Data				
LR	0.690	[0.72, 0.67]	[0.64, 0.74]	[0.68, 0.70]
KNN	0.982	[1, 0.97]	[0.97, 1]	[0.98, 0.98]
Decision tree	0.941	[0.97, 0.91]	[0.91, 0.97]	[0.94, 0.94]

LR- Logistic Regression, KNN- K Nearest Neighbor

The results of the implementation of both original and re-balanced data (SMOTE) for all the three algorithms (Logistic regression, KNN, and Decision tree) are shown in Table 1. From the table, it is clear that although the accuracy of all three models on original data is optimal however the overall performance in terms of precision, accuracy, and F1 score is poor. This is because the data is highly imbalanced and is only able to predict stars with no exoplanets correctly and fails to predict stars with exoplanets. Evidently, such models cannot be used for prediction since accuracy alone is not a true indicator of model performance for classification tasks. Understanding the importance of a balanced dataset and classes, we considered re-balancing the data, as discussed in the previous section. After re-balancing, improvements in the overall performance of models were observed for all three algorithms. Although the observed accuracy is lesser than that observed in the original

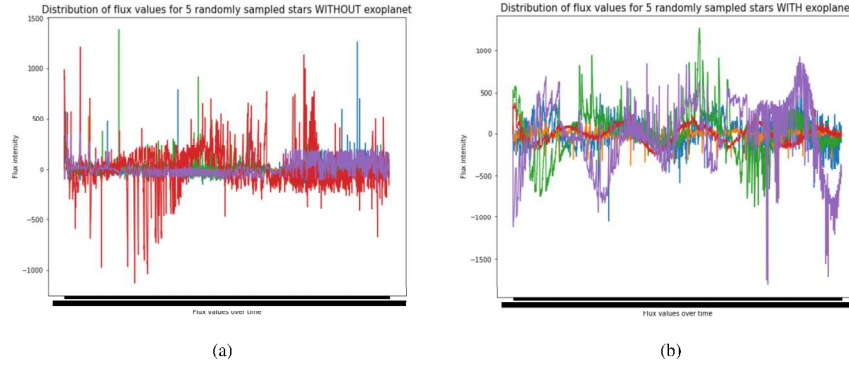


Fig. 1. Distribution of light intensity for 5 randomly observed sampled stars (a) without exoplanets and (b) with exoplanets.

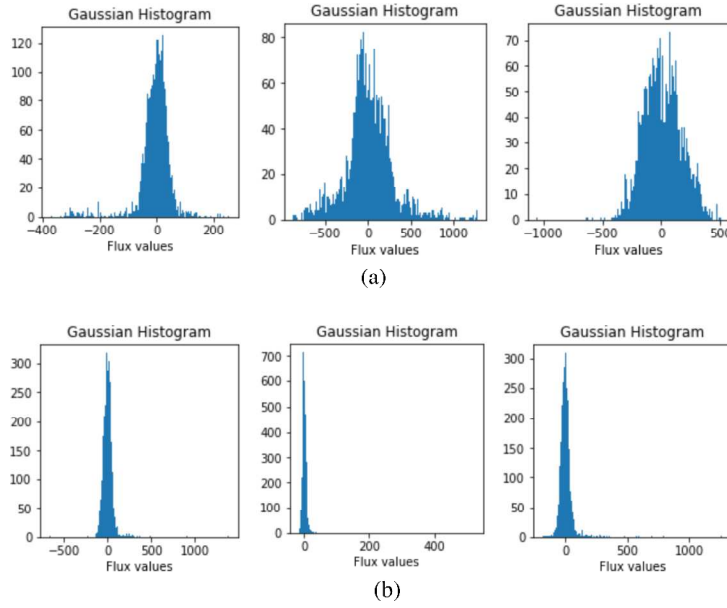


Fig. 2. Gaussian distribution of flux values observed over time for (a) stars with exoplanets and (b) stars without exoplanets

dataset, still the overall performance is better considering the values of precision, recall, and F1 score. The best performance was given by K-Nearest Neighbor with an accuracy of 98.2% and an F1 score of 98%, followed by a Decision tree and logistic regression.

To further dive into performance of the best performing KNN model over SMOTE data, we considered plotting a Receiver Operating Characteristics curve with it's Area under the curve (Fig 3). The curve shows the performance of a classification model at all classification thresholds. AUC (the degree or measure of separability) for the curve was found to be 99.40%. The figure confirms the optimal working of the model for predicting stars with and without exoplanets.

V. PREDICTION OVER LIMITED DATA

One of the crucial point of concern of this method for detection of exoplanets is the availability of data. The KEDC3 data used in this research consisted of 3197 observations of light intensity for each star. This is often difficult to observe. Thus, we considered implementing the best performing KNN model with limited data i.e. with limited observation of light intensity. We implemented this in multiple thresholds of data limitation with 75%, 50%, 25%, 10%, 5% and 1% of the balanced KEDC3 data. The results of the same has been shown in the Table II. The trend of decreasing performance of the model with limiting data size can also be seen in the Figure 4. However, even with decline in performance (90.1% accuracy for the model) with only 1% data i.e.31 observations of the light intensity is considerable enough for appropriate detection.

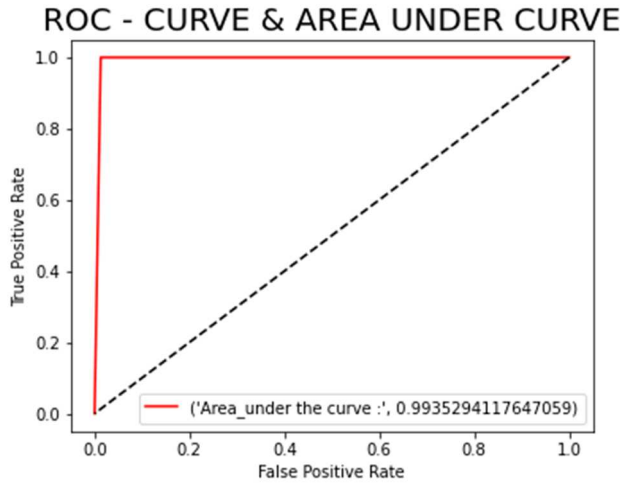


Fig. 3. Area under the ROC curve for KNN (SMOTE) model

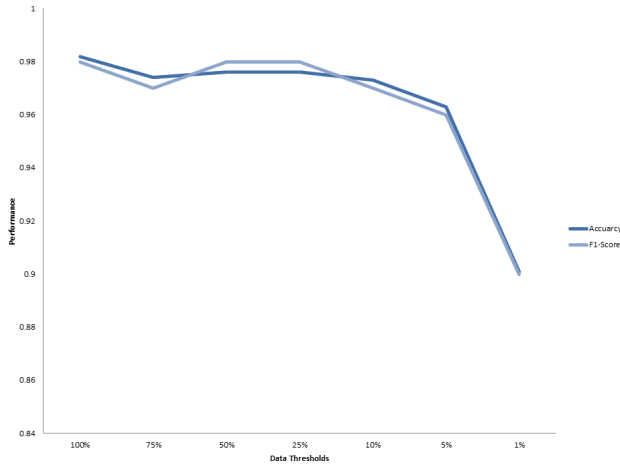


Fig. 4. Trend of model performance over limiting data size

TABLE II
PERFORMANCE OF KNN MODEL ON LIMITED DATASET

	Accuracy	Precision	Recall	F1-Score
100%	0.982	[1, 0.97]	[0.97, 1]	[0.98, 0.98]
75%	0.974	[1, 0.95]	[0.95, 1]	[0.97, 0.97]
50%	0.976	[1, 0.95]	[0.95, 1]	[0.98, 0.98]
25%	0.976	[1, 0.96]	[0.96, 1]	[0.98, 0.98]
10%	0.973	[1, 0.95]	[0.95, 1]	[0.97, 0.97]
5%	0.963	[1, 0.94]	[0.93, 1]	[0.96, 0.96]
1%	0.901	[0.97, 0.85]	[0.83, 0.97]	[0.90, 0.91]

VI. CONCLUSION

Research in astronomy and space science has a very wider scope. With more data getting collected by the modern methods, there is void for machine learning to implemented for

analysis and development. This paper focused on using various classification algorithms for detection of exoplanets using the light intensity of the stars which are plausibly the ones that the target exoplanets orbit around. Out of multiple algorithms used, K-Nearest Neighbor was found to be the best performing. We also considered using Synthetic Minority Oversampling Technique (SMOTE) for re-balancing the data for better results. Further, we also considered implementing the KNN algorithm when the available data is reduced than original. We considered thresholds for limiting the data for the said experiment. The key results of this research are as follow:

- K-Nearest Neighbor found to be best performing with overall accuracy of 98.20% and F1-score of 98%.
- Model with data limited to only 1%, still gave an accuracy of 90.1% and F1-score of 90%.
- Overall results of the research were found suitable for detection of exoplanets in real-time.

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